
MECH.5200 NUMERICAL METHODS FOR PDES
HOMEWORK # 5: AI-ASSISTED PDE SOLVING
WITH MATLAB PDE TOOLBOX
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1 Description

Throughout this course you have solved PDEs using Finite Differences, Finite Elements, Finite Volumes, and Physics-Informed Neural Networks. In this final homework, you will use a **commercial-grade PDE solver** — the MATLAB Partial Differential Equation Toolbox — and your **new collaborator**: Generative AI.

Important — Grading Philosophy: The primary goal of this assignment is to **document and demonstrate your effective use of AI as an engineering tool**. A completed, fully working solution is ideal, but **any serious, well-documented attempt is sufficient for full credit**. We care far more about the quality of your AI interaction — the prompts you crafted, the iterations you performed, the errors you encountered and fixed, and your reflections on the process — than about a perfect final result.

Expected time: approximately **1–1.5 hours** for Parts 2 and 3 (after completing the training in Part 1).

Tools you may use: GitHub Copilot, ChatGPT, Claude, Gemini, or any other Generative AI tool. You must identify which tool(s) you used.

2 Part 1: MathWorks Training (20 points)

Complete the free, self-paced MathWorks online course:

Introduction to Finite Element Analysis with MATLAB
[https://matlabacademy.mathworks.com/details/
introduction-to-finite-element-analysis-with-matlab/otmlfea](https://matlabacademy.mathworks.com/details/introduction-to-finite-element-analysis-with-matlab/otmlfea)

This course (~1 hour) teaches you to:

- Create and import 2D/3D geometries in PDE Toolbox
- Generate and visualize finite element meshes

- Define material properties, boundary conditions, and loads
- Solve PDEs and analyze results

Deliverable: Submit a screenshot or PDF of your **course completion certificate** from MATLAB Academy.

3 Part 2: PDE Toolbox Project (60 points)

Choose **one** of the six problems below. Using a Generative AI tool of your choice, write MATLAB code that uses the PDE Toolbox to solve the problem. **All code must be generated by AI** — you may (and should) iterate, debug, and refine the AI's output, but the starting point for every piece of code should be an AI-generated response.

3.1 Problem Options

Problem 1: Heat Sink with Fins (*Thermal*)

Model steady-state heat conduction in a 2D cross-section of a heat sink with rectangular fins. Apply a uniform heat flux at the base (representing contact with a hot chip) and convective cooling (Robin boundary condition) on all fin surfaces exposed to air.

Parameter study: Vary the number of fins or fin length and plot the maximum base temperature as a function of the parameter.

Problem 2: Bracket Under Load (*Structural*)

Perform a 2D plane-stress analysis of an L-shaped or T-shaped bracket. Fix one face (Dirichlet BC) and apply a distributed load on another face.

Parameter study: Vary the applied load magnitude or bracket thickness and plot the maximum von Mises stress and displacement.

Problem 3: Engine Component Thermal Stress (*Coupled Thermal-Structural*)

First solve steady-state heat conduction in a simplified 2D engine block cross-section with an internal heat source (combustion chamber). Then use the resulting temperature field as input for a thermal stress analysis.

Parameter study: Vary the heat source intensity and plot the peak thermal stress.

Problem 4: Diffusion-Reaction in a Catalyst Pellet (*Chemical*)

Solve the steady-state diffusion-reaction equation $D\nabla^2 c = k \cdot c$ inside a 2D circular pellet cross-section. Apply a fixed concentration on the outer boundary (Dirichlet BC).

Parameter study: Vary the Thiele modulus $\phi = R\sqrt{k/D}$ (by changing k) and plot the radial concentration profile for several values of ϕ .

Problem 5: Tissue Ablation Heat Distribution (*Biomedical*)

Solve the transient heat equation in a 2D tissue cross-section with a localized Gaussian heat source representing a laser or radiofrequency probe. Use biological tissue thermal properties ($k \approx 0.5 \text{ W}/(\text{m}\cdot\text{K})$, $\rho \approx 1000 \text{ kg}/\text{m}^3$, $c_p \approx 4000 \text{ J}/(\text{kg}\cdot\text{K})$).

Parameter study: Vary the source power or duration and plot the temperature evolution at the center. Identify the region exceeding the ablation threshold ($\sim 60^\circ\text{C}$).

Problem 6: Femur Bone Cross-Section Under Load (*Biomedical / Structural*)

Perform a 2D plane-stress analysis of a simplified femur mid-shaft cross-section, modeled as a hollow elliptical geometry (cortical bone surrounding a medullary cavity). Apply a compressive body-weight load.

Parameter study: Vary the cortical wall thickness (simulating osteoporosis progression) and plot the peak stress as a function of wall thickness.

3.2 Requirements

Your code and analysis must include the following steps (regardless of which problem you choose):

1. **Geometry:** Define or import the 2D geometry using PDE Toolbox functions.
2. **Mesh:** Generate a finite element mesh. Show a mesh plot.
3. **Material properties & BCs:** Set appropriate material properties, boundary conditions, and loads/sources.
4. **Solution:** Solve the PDE and create at least one visualization of the result (e.g., contour plot, surface plot).
5. **Parameter study:** Vary one parameter as described in your chosen problem. Present results in a clear plot.
6. **AI conversation log:** Include the **complete** AI conversation (prompts and responses) as an appendix. You may use screenshots, exported chat logs, or copy-paste.

3.3 Grading Breakdown

Category	Points
AI process documentation & prompt quality	20
Geometry + mesh setup	10
Solution attempt & visualization	10
Parameter study attempt	10
Reflection on AI strengths/limitations encountered	10
Total	60

The **AI process documentation** category is weighted most heavily. We are looking for:

- Clear, well-crafted prompts that provide context (the PDE, the toolbox, variable names, etc.)
- Evidence of iteration — how did you refine your prompts when the first output wasn't right?
- Honest reporting of failures and how you worked around them

4 Part 3: AI-Generated Report (20 points)

Write a short technical report documenting your project. The report itself must be **generated using a Generative AI tool** (you may use any format: $\text{\LaTeX}$, Markdown, or HTML). You should prompt the AI to produce the report structure and content, then edit for accuracy.

The report must include:

1. **Title and introduction** — state the problem and its engineering relevance
2. **Methodology** — describe the PDE, geometry, BCs, and numerical approach
3. **Results** — include figures (mesh, solution, parameter study) with captions
4. **Discussion** — interpret the results physically; comment on mesh sensitivity if applicable
5. **Conclusion** — summarize findings
6. **AI usage acknowledgment** — which tool(s) you used and a brief description of your prompting process for the report

Category	Points
Completeness (all sections present)	5
Quality of figures (labeled axes, captions, readable)	5
Clarity of writing	5
AI process documentation for the report	5
Total	20

5 Bonus (+10 points)

Earn up to 10 bonus points by completing one or both of the following:

1. **3D Extension (+5 pts):** Extend your chosen problem to a 3D geometry using PDE Toolbox. Show the 3D mesh and solution. Briefly discuss what changed in the AI prompting process for 3D vs. 2D.
2. **Analytical/Hand-Coded Comparison (+5 pts):** Compare your AI-generated PDE Toolbox solution to either (a) an analytical solution (if one exists for your problem) or (b) a hand-coded FEM/FD solver from earlier in the course. Discuss accuracy, time-to-solution, and ease of implementation.

Summary of Deliverables

1. Course completion certificate (Part 1)
2. MATLAB code — all .m files (Part 2)
3. AI conversation log — screenshots or exported chat (Part 2, Appendix)
4. Technical report in your chosen format (Part 3)
5. Bonus materials, if applicable

Good luck, and have fun exploring what AI can (and can't) do!